

WEBB CITY JOPLIN, MO, USA

WMO: 723495

Lat: 37.152N

Lon: 94.495W

Elev: 972

StdP: 14.19

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13987

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	13.7	-1.0	5.4	11.3	3.7	6.9	17.2	27.9	53.5	25.9	51.9	8.2	10	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	97.3	75.1	94.3	75.6	91.6	75.2	78.9	90.5	77.9	89.4	76.9	88.0	9.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	139.1	85.7	74.5	133.6	84.5	73.3	128.4	83.2	43.1	90.8	42.0	89.7	40.9	87.9	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	22.1	19.5	1.4	100.4	5.2	3.4	-2.3	102.8	-5.3	104.8	-8.2	106.8	-11.9	109.3
			0.5	80.9	4.9	1.9	-3.0	82.2	-5.9	83.3	-8.6	84.4	-12.2	85.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	35.4	39.7	48.8	58.4	67.0	76.0	80.2	79.5	71.4	60.0	48.6	38.5
	DBStd	18.15	11.99	11.87	11.36	9.32	7.84	5.67	5.37	5.68	7.65	9.30	10.23	10.80
	HDD50	1534	472	325	160	29	2	0	0	0	0	21	145	380
	HDD65	4010	918	709	511	237	73	4	0	1	28	211	497	821
	CDD50	4720	19	37	123	282	529	779	937	914	641	331	104	24
	CDD65	1717	0	1	9	39	135	332	472	450	219	55	5	0
	CDH74	17531	0	9	68	320	1118	3286	5343	4897	2048	412	30	0
	CDH80	7586	0	1	7	58	291	1327	2587	2414	805	94	2	0
Wind	WSAvg	9.2	9.9	10.1	10.6	11.2	9.3	8.8	7.7	7.3	7.4	8.7	9.8	9.5
Precipitation	PrecAvg	43.6	1.7	2.2	3.5	4.1	5.0	5.6	3.2	3.8	5.0	3.6	3.5	2.9
	PrecMax	65.2	4.1	7.8	9.1	10.4	12.9	12.2	12.4	8.5	19.1	9.1	12.3	7.7
	PrecMin	18.3	0.0	0.2	0.8	0.1	1.5	0.8	0.2	0.0	1.0	0.0	0.0	0.5
	PrecStd	9.9	1.2	1.6	1.9	2.2	2.6	2.6	2.6	2.3	4.0	2.3	2.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.6	74.7	80.2	85.3	89.2	96.0	101.0	101.7	96.9	87.0	78.1	69.9
		MCWB	57.6	60.1	62.2	65.5	72.3	75.5	74.0	75.1	72.9	71.0	63.2	60.0
	2%	DB	63.9	68.3	75.4	81.2	86.1	92.2	97.1	97.7	91.0	82.5	72.6	64.7
		MCWB	53.8	56.0	60.6	64.3	71.3	75.6	75.4	75.1	73.1	67.2	60.4	56.1
	5%	DB	59.5	63.8	71.0	77.7	83.4	89.9	94.2	94.5	88.1	78.7	68.9	59.9
		MCWB	50.7	54.1	57.8	62.6	70.4	74.8	76.5	75.1	72.7	64.9	59.4	52.3
	10%	DB	53.8	59.1	67.0	74.2	80.6	87.4	91.4	91.5	84.7	74.9	64.7	55.2
		MCWB	46.4	49.7	56.4	61.5	69.0	74.1	76.2	74.6	70.7	64.3	56.1	48.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.1	63.2	66.0	69.5	75.5	79.4	80.6	80.0	77.3	73.3	66.5	62.4
		MCDB	66.6	69.3	75.2	79.8	84.8	91.0	91.8	91.5	88.3	82.8	73.1	68.1
	2%	WB	56.8	59.7	63.0	67.5	73.7	77.7	79.3	78.3	75.4	70.8	63.4	58.5
		MCDB	62.1	66.5	71.9	76.6	82.9	88.6	90.8	90.7	87.2	78.4	69.9	63.5
	5%	WB	51.3	55.3	60.4	65.7	72.1	76.4	78.3	77.2	74.2	68.5	60.4	53.8
		MCDB	57.8	61.4	69.0	74.2	81.0	86.8	89.6	89.7	85.0	74.9	66.6	58.3
	10%	WB	46.3	50.4	57.5	63.5	70.2	75.0	77.3	75.9	72.6	66.1	57.6	49.4
		MCDB	53.2	58.6	66.0	71.8	78.3	85.0	88.4	87.7	82.0	73.0	63.7	54.3
Mean Daily Temperature Range	5% DB	MDBR	18.8	19.7	20.8	21.0	19.4	19.0	19.3	20.5	21.0	21.1	20.0	17.9
		MCDBR	24.9	26.2	25.3	24.0	21.0	20.6	21.9	23.4	22.6	23.2	23.3	23.2
		MCWBR	16.2	17.2	14.5	13.0	10.4	8.7	7.2	7.6	9.0	11.3	14.6	16.0
	5% WB	MCDBR	22.1	23.4	21.0	19.5	18.9	19.0	18.8	19.6	19.7	18.0	21.0	20.4
		MCWBR	16.8	18.0	14.1	12.5	10.4	9.1	8.1	8.1	8.6	10.3	15.4	16.3
Clear-Sky Solar Irradiance	taub	0.304	0.313	0.346	0.380	0.423	0.423	0.440	0.443	0.387	0.345	0.311	0.291	
	taud	2.449	2.437	2.388	2.313	2.250	2.280	2.277	2.246	2.369	2.464	2.502	2.507	
	Ebn at Noon	281	293	291	285	273	272	267	264	274	277	277	279	
	Edh at Noon	27	30	35	40	43	42	42	43	36	30	25	23	
All-Sky Solar Radiation	RadAvg	762	947	1261	1600	1793	2060	2048	1841	1560	1140	814	648	
	RadStd	68	142	80	104	148	168	127	120	116	138	89	76	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page